

TIME OF DAY ANALYSIS (24-HOUR CLOCK)

Dataset: 30-day attack logs (84 events)

Analyst: Raul Austero Varias Jr.

Section A: Hourly Event Count

Hour (UTC)	Count	Percentage	Cumulative %
00	0	0.0%	0.0%
01	0	0.0%	0.0%
02	59	70.2%	70.2%
03	11	13.1%	83.3%
04	0	0.0%	83.3%
05	0	0.0%	83.3%
06	0	0.0%	83.3%
07	0	0.0%	83.3%
08	0	0.0%	83.3%
09	0	0.0%	83.3%
10	0	0.0%	83.3%
11	0	0.0%	83.3%
12	0	0.0%	83.3%
13	0	0.0%	83.3%
14	14	16.7%	100.0%
15	0	0.0%	100.0%
16	0	0.0%	100.0%
17	0	0.0%	100.0%
18	0	0.0%	100.0%
19	0	0.0%	100.0%
20	0	0.0%	100.0%
21	0	0.0%	100.0%
22	0	0.0%	100.0%
23	0	0.0%	100.0%
Total	84	100.0%	100.0%

Section B: Top & Lowest Activity Windows

Top 3 Active Hours:

1. 02:00 - 02:59 UTC (59 events, 70.2%)
2. 03:00 - 03:59 UTC (11 events, 13.1%)
3. 14:00 - 14:59 UTC (14 events, 16.7%)

Lowest 3 Active Hours: Any hours between 00:00–01:59, 04:00–13:59, or 15:00–23:59 (0 events, 0.0%).

Section C: Attack Phase by Time of Day

Attack Phase	Most Frequent Hour	Count
Initial Access	14:00–14:59	4 (log 1, 21, 54, 83)
Execution	14:00–14:59	7 (log 2, 3, 22, 23, 55, 56, 84)
Persistence	14:00–14:59	3 (log 4, 24, 57)
Discovery	02:00–02:59	14 (log 5, 12, 17, 25, 31, 36, 40, 45, 50, 58, 64, 69, 73, 78)
Credential Access	02:00–02:59	21 (log 6, 7, 13, 18, 19, 26, 27, 32, 37, 38, 41, 46, 51, 52, 59, 60, 65, 70, 71, 74, 79)
Lateral Movement	02:00–02:59	10 (log 8, 14, 28, 33, 42, 47, 61, 66, 75, 80)
Collection	02:00–02:59	10 (log 9, 15, 29, 34, 43, 48, 62, 67, 76, 81)
C2	02:00–02:59	4 + 1 (at 03:00) (log 10, 20, 39, 53, 72)
Exfiltration	03:00–03:59	10 (log 11, 16, 30, 35, 44, 49, 63, 68, 77, 82)

Section D: Time of Day Insights

Business hours (08:00-17:00): **14/84 = 16.7%**

Off-hours (00:00-07:00, 18:00-23:00): **70/84 = 83.3%**

The adversary operates almost entirely during off-hours - 83.3% of all activity is packed into just two hours (02:00-03:59 UTC). The much smaller 16.7% business-hours window is used exclusively for phishing delivery, execution, and persistence; it never contains discovery, credential theft, lateral movement, collection, or exfiltration. This split points to a deliberate strategy: compromise during the workday when a phishing email is most likely to be opened and clicked, then wait and carry out all noisy activity overnight, when SOC staffing and defender visibility are typically lowest.

TOOL CHOICE ANALYSIS

Dataset: 30-day attack logs (84 events)

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Section A: Tool Frequency

Tool Name	Exact Log IDs	Count	Percentage	Rank
PowerShell	3, 6, 9, 15, 18, 23, 26, 29, 34, 37, 43, 48, 51, 56, 59, 62, 67, 70, 76, 81	20	23.8%	1
cmd.exe	5, 12, 17, 25, 31, 36, 40, 45, 50, 58, 64, 69, 73, 78	14	16.7%	2
Mimikatz	7, 13, 19, 27, 32, 38, 41, 46, 52, 60, 65, 71, 74, 79	14	16.7%	2
Psexec	8, 14, 28, 33, 42, 47, 61, 66, 75, 80	10	11.9%	4
HTTPS POST (Exfil)	11, 16, 30, 35, 44, 49, 63, 68, 77, 82	10	11.9%	4
HTTPS Beacon (C2)	10, 20, 39, 53, 72	5	6.0%	6
Phishing Email	1, 21, 54, 83	4	4.8%	7
User Action (click)	2, 22, 55, 84	4	4.8%	7
Scheduled Task	4, 57	2	2.4%	9
Registry Run Key	24	1	1.2%	10
Total		84	100.0%	

Section B: Tool by Attack Phase

Attack Phase	Primary Tool	Secondary Tool	Tertiary Tool
Initial Access	Phishing Email		
Execution	User Action (click)	PowerShell	
Persistence	Scheduled Task	Registry Run Key	
Discovery	cmd.exe		
Credential Access	Mimikatz	PowerShell	
Lateral Movement	Psexec		
Collection	PowerShell		
Command & Control	HTTPS Beacon		
Exfiltration	HTTPS POST		

Section C: Tool Sequence Analysis

Sequence 1: cmd.exe → Mimikatz → PsExec | no PowerShell download step (occurs 7 times on 03-28, 04-04, 04-09, 04-11, 04-18, 04-22, 04-23).

This is actually the more common pattern, not the PowerShell variant.

Sequence 2: cmd.exe → PowerShell → Mimikatz → PsExec | PowerShell downloads Mimikatz (occurs 3 times on 03-25/26, 04-01/02, 04-16)

Sequence 3: PowerShell (Collection) → HTTPS POST | Exfiltration (occurs 10 times)

Collection stage, every incident that reaches exfiltration.

Section D: Tool Choice Insights

- Most used tool overall: PowerShell → 20 occurrences, 23.8%
- Most diverse phase (most different tools): Execution, Persistence, and Credential Access each use exactly 2 distinct tools.
- Most tool-consistent phases (only 1-2 tools): Discovery, Lateral Movement, Collection, and Exfiltration.

Describe the adversary's tool preference pattern (3-5 sentences):

The adversary maintains strict tool consistency for key phases, exclusively using cmd.exe for Discovery, PsExec for Lateral Movement, and HTTPS POST for Exfiltration. However, Credential Access shows operational flexibility, as the actor alternates between re-downloading Mimikatz via PowerShell and using pre-staged binaries. This variance indicates the adversary follows a flexible, multi-branch playbook rather than a single rigid execution sequence.

TOOL-TEMPORAL CORRELATION MATRIX

Hours with highest tool concentration

Tool	Peak Activity Hour	Off-Hours Activity	Business Hours Activity
PowerShell	02:00-02:59	15% (3 at hour 14, Execution)	85%
cmd.exe	02:00-02:59	0%	100%
Mimikatz	02:00-02:59	0%	100%
PsExec	02:00-02:59	0%	100%

Phishing Email	14:00-14:59	100%	0%
User Action	14:00-14:59	100%	0%
HTTPS Beacon	02:00-02:59	0%	100%
Exfiltration	03:00-03:59	0%	100%

Which tools are STRICTLY used during off-hours? cmd.exe, Mimikatz, PsExec, HTTPS Beacon, HTTPS POST

Which tools are used during BOTH off-hours and business hours? PowerShell - the only tool spanning both windows, since it's used for both the initial payload execution and post-exploitation work.

What is the most predictable tool-hour combination? Exfiltration via HTTPS POST at 03:00-03:59 UTC

What does this suggest about the adversary's operational security? This suggests that the adversary has weak operational security because their activities follow a very consistent schedule. They do not appear to change their timing enough to avoid pattern-based detection. This makes their behavior predictable and easier for defenders to detect.

ATTACK PROGRESSION TIMING

Section A: From Initial Access to Discovery

03-25→26: 14:22:15 → 02:15:33 = 11h 53m

04-01→02: 14:15:22 → 02:15:33 = 12h 0m

04-15→16: 14:18:22 → 02:12:33 = 11h 54m

Average: 11h 56m (12 hours)

Section B: Discovery → Credential Access

03-26: 02:15:33 → 02:22:18 = 6m 45s

04-02: 02:15:33 → 02:22:05 = 6m 32s

04-16: 02:12:33 → 02:20:44 = 8m 11s

Average: 7 minutes

Section C: Credential Access → Lateral Movement

03-26: 02:35:44 → 02:36:10 = 26 seconds

04-02: 02:30:44 → 02:35:44 = 5 minutes

04-16: 02:28:11 → 02:35:44 = 7m 33s

Average: 4 minutes

Section D: Collection → Exfiltration

03-25/26: 02:45:22 → 03:15:33 = 30 minutes

03-28: 02:40:10 → 03:05:22 = 25 minutes

04-02: 02:45:10 → 03:00:22 = 15 minutes

04-04: 02:50:22 → 03:10:44 = 20 minutes

Average: 22.5 minutes

Section E: Progression Summary

Phase	Typical Time (UTC)	Duration from Previous
Initial Access	14:20	N/A
Execution	14:21	1 min after IA
Persistence	14:22	1 min after Exec
Discovery	02:15 (next day)	12 hours after IA
Credential Access	02:22	7 min after Disc
Lateral Movement	02:29	4 min after CA
Collection	02:45	15-19 min after LM
Exfiltration	03:07	22-23 min after Coll

ATTACK PROGRESSION TIMING

Dataset: 30-day attack logs (84 events)

Analyst: Raul Austero Varias Jr.

Section A: Time Between Phases

Incident 2026-03-25: Initial Access 14:22:15 → Discovery next day 02:15:33 → Delta = **11 hours 53 minutes**

Incident 2026-04-01: Initial Access 14:15:22 → Discovery next day 02:15:33 → Delta = **12 hours 0 minutes**

Incident 2026-04-15: Initial Access 14:18:22 → Discovery next day 02:12:33 → Delta = **11 hours 54 minutes**

Incident 2026-04-23: Initial Access 14:22:10 → Discovery = approximately **11 hours 56 minutes** - expected around 02:18 UTC on 04-24, but not present in the dataset.

AVERAGE time from Initial Access to Discovery: 11 hours 56 minutes (12 hours)

Section B: Time Between Discovery and Credential Access

Incident 2026-03-25: Discovery 02:15:33 → Credential Access 02:22:18 → Delta = **6 minutes 45 seconds**

Incident 2026-04-01: Discovery 02:15:33 → Credential Access 02:22:05 → Delta = **6 minutes 32 seconds**

Incident 2026-04-15: Discovery 02:12:33 → Credential Access 02:20:44 → Delta = **8 minutes 11 seconds**

AVERAGE time from Discovery to Credential Access: 7 minutes

Section C: Time Between Credential Access and Lateral Movement

Incident 2026-03-25: Credential Access (Mimikatz) 02:35:44 → Lateral Movement 02:36:10 → Delta = **26 seconds**

Incident 2026-04-01: Credential Access (Mimikatz) 02:30:44 → Lateral Movement 02:35:44 → Delta = **5 minutes**

Incident 2026-04-15: Credential Access (Mimikatz) 02:28:11 → Lateral Movement 02:35:44 → Delta = **7 minutes 33 seconds**

AVERAGE time from Credential Access to Lateral Movement: ~4 minutes

Section D: Time Between Collection and Exfiltration

Incident 2026-03-25: Collection 02:45:22 → Exfiltration 03:15:33 → Delta = **30 minutes 11 seconds**

Incident 2026-03-28: Collection 02:40:10 → Exfiltration 03:05:22 → Delta = **25 minutes 12 seconds**

Incident 2026-04-02: Collection 02:45:10 → Exfiltration 03:00:22 → Delta = **15 minutes 12 seconds**

Incident 2026-04-04: Collection 02:50:22 → Exfiltration 03:10:44 → Delta = **20 minutes 22 seconds**

AVERAGE time from Collection to Exfiltration: 22.5 minutes

Section E: Progression Summary

Phase	Typical Time UTC	Duration from Previous
Initial Access	14:19	N/A
Execution	14:20	1 minute after AI
Persistence	14:22	1-2 minutes after Exec
Discovery	02:15 (next day)	12hrs after AI
Credential Access	02:13	7 minutes after Discovery
Lateral Movement	02:26	4 minutes after Credential Access
Collection	02:45	10–19 minutes after Lateral Movement
Exfiltration	03:07	22-23 minutes after Collection

Section F: Predictive Statement

1. If a new phishing email is sent at 14:20 UTC on a business day, when will the adversary likely begin Discovery?

If a new phishing email is sent at 14:20 UTC on a business day, the adversary will likely begin Discovery at approximately 02:15 UTC the following day — about 12 hours later.

2. What tools will the adversary use in the first 30 minutes after successful Initial Access?

In the first 30 minutes after successful Initial Access, the adversary will use User Action (double-click) to PowerShell (payload download) to Scheduled Task or Registry Run Key (persistence) all occurring within roughly 1-2 minutes of the initial click.

3. What is the adversary's most predictable behavior (single highest-confidence pattern)?

The adversary's most predictable behavior is Discovery timing across all 14 observed incidents, Discovery via cmd.exe net commands occurs within the same narrow 02:12-02:15 UTC window every single time, with zero exceptions making it the single most reliable early-warning indicator in the dataset.

Full predictive intelligence statement:

The SilentBanker adversary operates on a highly predictable 12-hour cycle initial compromise happens during business hours (14:15-14:23 UTC) via phishing, and post-exploitation activity begins almost exactly 12 hours later, always inside the 02:00-03:59 UTC window. Discovery consistently starts around 02:12-02:15 UTC using cmd.exe net commands, followed roughly 7 minutes later by Credential Access via Mimikatz. When lateral movement follows, it occurs within about 4 minutes of credential theft, and when exfiltration follows collection, it takes an average of 22-23 minutes to complete. This rigid timing, repeated across 30 days with almost no variation, means defenders can predict the adversary's next move with high confidence once a phishing click is detected. The highest-value detection window is 02:00-03:30 UTC, and the single most reliable trigger for an alert is any net group/net view/net user command executed from cmd.exe during that window on a host that received a suspicious email roughly 12 hours earlier.

PATTERN SUMMARY REPORT

Adversary: SilentBanker Operator

Dataset Period: 2026-03-25 to 2026-04-23 (30 days)

Analyst: Raul Austero Varias Jr.

Date: September 20, 2026

EXECUTIVE SUMMARY

The SilentBanker operator shows an extremely disciplined, near-mechanical operating pattern across the 30-day dataset of 84 confirmed malicious events. 83.3% of all activity is concentrated in a single overnight window (02:00-03:59 UTC), triggered almost exactly 12 hours after a business-hours phishing compromise at 14:00-14:59 UTC. PowerShell is the adversary's most-used and most versatile tool (23.8% of all events), appearing across three separate attack phases, while Discovery, Lateral Movement, and Exfiltration each rely on a single dedicated tool with zero deviation across all 30 days. Rather than following one single fixed sequence, the actor runs at least two distinct post-Discovery playbook branches a full data-theft path (lateral movement and exfiltration) and a shorter credential-only path that stops after beaconing to C2. This level of consistency suggests a scripted or automated operation rather than manual, improvised tradecraft.

TEMPORAL PATTERNS

Primary Activity Window: 02:00 - 04:00 UTC

Secondary Activity Window: 14:00 - 15:00 UTC

Notable Temporal Finding #1: 70.2% of all 84 events (59 events) occur within a single hour — 02:00-02:59 UTC — the highest concentration of activity in any hour of the dataset.

Notable Temporal Finding #2: Business-hours activity (16.7% of events) is strictly limited to the first three phases of the attack chain Initial Access, Execution, and Persistence. Across all 14 observed incidents, not a single Discovery, Credential Access, Lateral Movement, Collection, C2, or Exfiltration event ever occurs during business hours.

TOOL CHOICE PATTERNS

Primary Tool Set: PowerShell (23.8%), cmd.exe (16.7%), Mimikatz (16.7%)

Tool-Phase Mapping:

Discovery: cmd.exe (always) → 100% of Discovery events

Credential Access: Mimikatz (always) → 100% of Credential Access events

Lateral Movement: PsExec (always) → 100% of Lateral Movement events

Tool Sequence (most common chain):

Phishing Email → User Action → PowerShell → Scheduled Task/Registry Run Key (Day 1 Staging) → cmd.exe → Mimikatz → PsExec → PowerShell → HTTPS Beacon → HTTPS POST (Day 2 Execution & Exfil)

TOOL-TEMPORAL CORRELATION

Tool that NEVER appears during business hours: cmd.exe

Tool that ALWAYS appears during off-hours: HTTPS POST (Exfiltration)

Tool that appears at ANY time (no temporal restriction): PowerShell the only tool used during both business hours and off-hours.

PREDICTIVE INTELLIGENCE

Next Likely Action:

Time: The night following any detected phishing click, approximately 02:00-02:15 UTC

Tool(s): cmd.exe, Mimikatz, and PsExec (if lateral movement follows)

Target: The same compromised workstation first, then internal file servers if the attack progresses to lateral movement

Detection Recommendation:

Best detection window: 02:00-03:30 UTC

Log sources to monitor during this window: EDR/endpoint process logs (for cmd.exe and Mimikatz execution), network/firewall logs (for outbound HTTPS to known C2 IP ranges), and SIEM correlation alerts tied to the initial phishing host

Suspicious tool combinations to alert on: cmd.exe net group/net view/net user commands followed within 10 minutes by Mimikatz process activity; PowerShell downloading an executable followed by a scheduled task or registry Run key creation

MITRE ATT&CK MAPPING (Patterns Identified)

Pattern	MITRE Tactic	MITRE Technique	Technique ID
Off-hour Discovery	Discovery	Remote System Discovery	T1018
PowerShell + Mimikatz combo	Credential Access	OS Credential Dumping: LSASS Memory	T1003.001
Psexec lateral movement	Lateral Movement	Remote Services: SMB/Windows Admin Shares	T1021.002
Scheduled Task persistence	Persistence	Scheduled Task/Job: Scheduled Task	T1053.005
Compressed archive before exfil	Collection	Archive Collected Data: Archive via Utility	T1560.001
HTTPS beacon disguised as normal traffic	Command and Control	Application Layer Protocol: Web Protocols	T1071.001

APPENDIX: RAW DATA SUMMARY

Total Events Analyzed: 84

Total Unique Tools: 10

Total Distinct Attack Phases: 9

Peak Activity Hour: 02:00-02:59 UTC, with 59 events

Most Used Tool: PowerShell, with 20 appearances